

Value-Grounded AI Alignment: Canonical Axiology, Normative Late Binding, and Extrinsic Assurance

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Scientific narrative revision; synthetic shared-MLP mechanism-validation pilot

Abstract

Alignment methods for language models improve behavioral compliance through supervised fine-tuning, preference optimization, written principles, deliberation, and runtime controls. This paper studies whether explicit value and normative structure can participate in neural computation while volatile facts, contextual authority, and formal assurance remain separate. Value-Grounded AI Alignment (VGA) defines an external canonical axiology O_A^v , a derived neural module $A_\phi^v = C_\phi(O_A^v)$, late-bound normative sources, a domain ontology, a purpose model, and an evidence-bearing world state. The reference Value-Grounded Transformer Architecture (VGTA) uses these interfaces for representation, attention, routing, semantic state, and candidate-action assurance. A three-seed pilot compares A1, B, C1, and C2 shared-MLP proxies on a frozen synthetic benchmark with structural-OOD, conformance, coverage, calibration, and attack tests. The generated results provide mechanism-level evidence for the controlled scenario grammar and identify semantic-occlusion failures. The artifact contains no empirical Transformer study and does not establish moral truth, legitimate authority, formal safety, consciousness, or a solution to AI alignment.

Status. Version 0.4.0 scientific narrative revision; empirical protocol v0.3.0. Outputs are labeled `pilot-uncommitted` and describe a synthetic shared-MLP proxy, not a Transformer or population study. Claims remain limited to generated measurements and controls.

1 Introduction

Language-model alignment is commonly studied as a behavioral problem: training and evaluation move a model toward useful instruction following, preference satisfaction, or compliance with a written policy. That framing has produced effective methods, but it leaves open a narrower systems question. A model may satisfy a local behavioral target while losing track of which entities are affected, which values are at stake, which norms apply, what purpose governs the task, or whether current world-state assertions are reliable.

This paper asks whether those distinctions can become explicit computational inputs. The research question is:

Can a governed external canonical axiology be compiled into a derived neural module and used as structural conditioning, while contextual norms are late-bound, purpose selects domain relevance, world state remains evidence-bearing, and consequential candidate actions are independently checked?

The proposed answer is Value-Grounded AI Alignment (VGA), instantiated by the reference Value-Grounded Transformer Architecture (VGTA). VGA separates a canonical axiology O_A^v , normative assertions O_N , a domain ontology O_D , a purpose model P , and a time-indexed world state W_t . A compiler derives the non-authoritative module $A_\phi^v = C_\phi(O_A^v)$. Neural interventions may use that module for representation, attention, routing, or semantic state; normative authority and current evidence remain explicit context; and an independent verifier evaluates structured candidate actions before external effects.

The paper develops the model, places it among alignment, neuro-symbolic, knowledge-enhanced, and formal-methods research, and evaluates an initial mechanism-validation implementation. The implementation is a deterministic synthetic benchmark paired with a shared NumPy multilayer perceptron. It compares A1, B, C1, and C2 across three seeds, tests structural out-of-distribution behavior, candidate conformance, ontology coverage, calibration, and synthetic attacks, and generates all reported numerical artifacts from raw predictions.

1.1 Contributions

The contributions are:

1. a layered computational model distinguishing canonical axiology, normative authority, domain semantics, purpose relevance, uncertain world state, and extrinsic action assurance;
2. a reference architecture with separable semantic-to-neural compilation, structural conditioning, normative late binding, and a fixed-verifier boundary;
3. a falsifiable experimental design with matched model variants, structural-OOD splits, coverage degradation, security fixtures, candidate-action conformance, and seed-level uncertainty;
4. a runnable research artifact containing ontology fixtures, interface modules, a synthetic pilot, generated tables and figures, tests, and reproducibility records.

1.2 Scope

The manuscript evaluates whether explicit semantic structure is a useful inductive bias in a controlled proxy; it does not establish moral truth, legitimate authority, consciousness, machine moral status, formal safety, or a solution to AI alignment.

2 Background and Related Work

This section reviews behavioral alignment, explicit principles, neuro-symbolic reasoning, knowledge-enhanced Transformers, value pluralism, and formal constraint systems. It identifies a systems gap without treating any single tradition as a complete solution.

2.1 Contemporary language-model alignment

The Transformer introduced a sequence architecture based on self-attention rather than recurrence [Vaswani et al., 2017]. Modern alignment pipelines commonly combine pretrained language modeling with supervised examples, preference or reward optimization, policy specifications, and runtime restrictions. InstructGPT demonstrated that human demonstrations and rankings can improve instruction following [Ouyang et al., 2022]; preference learning established a general recipe for

learning from human comparisons [Christiano et al., 2017]. Constitutional and AI-feedback methods use written principles, self-critique, or model-generated feedback [Bai et al., 2022, Lee et al., 2023], while DPO changes the optimization path without removing dependence on preference data [Rafailov et al., 2023].

These methods expose distinct measurement problems. Preferences and rewards are proxies that may be overoptimized [Gao et al., 2023, Kim and Seo, 2024]. Correct specifications do not guarantee correct goals under distribution shift [Langosco et al., 2022, Shah et al., 2022]. Safety behavior can also change under adversarial prompts, fine-tuning, and reward tampering [Wei et al., 2023, Qi et al., 2023, Denison et al., 2024]. The present work treats behavioral alignment as a control condition and asks whether explicit semantic roles add measurable information beyond that condition.

2.2 Constitutional and deliberative alignment

Written principles and deliberation make policy reasoning more explicit. Constitutional methods use principles, self-critique, AI feedback, and preference optimization [Bai et al., 2022, Lee et al., 2023]. Deliberative alignment trains models to reason over safety specifications before responding [Guan et al., 2024]. These approaches provide close precedents for contextual normative reasoning. They need not, however, encode a separate distinction between what is value-bearing, what ought to be done, which authority governs, and which conflict remains unresolved. VGA uses these methods as compatible training or control choices rather than as substitutes for the layered representation.

2.3 Neuro-Symbolic Reasoning and Formal Constraint Systems

Neuro-symbolic AI combines statistical learning with symbolic representations, rules, logic, or structured inference. Surveys distinguish model-based and proof-based reasoning, learned versus supplied structure, and the degree of integration between neural and symbolic components [de Raedt et al., 2020, Marra et al., 2024, d’Avila Garcez et al., 2019]. DeepProbLog, Logic Tensor Networks, and differentiable rule induction connect logical or probabilistic structure to gradient-based learning [Manhaeve et al., 2018, Badreddine et al., 2022, Evans and Grefenstette, 2018].

Formal constraint systems provide a different role. Satisfiability Modulo Theories (SMT) asks whether a formula is satisfiable under combinations of background theories such as arithmetic, arrays, and uninterpreted functions [Barrett et al., 2021]. SMT-LIB defines shared theories, input languages, output conventions, and benchmark formats so that solvers and tools can be compared [Barrett et al., 2010, 2025]. Z3 is an SMT solver designed for efficient reasoning over such theories [de Moura and Bjørner, 2008]. These systems can establish properties of a formalized representation; they do not establish that the representation captures all relevant semantics or that a norm is legitimate.

Policy languages illustrate how this boundary is used in practice. The AWS Block Public Access work compiles a policy into a logical formula and uses SMT reasoning to check a trust-safety property [Bouchet et al., 2020]. The result is a policy-verification system with a defined abstraction and deployment boundary, not a proof that the underlying policy expresses every relevant human value. CSL-Core is one recent implementation example: a policy DSL is compiled to an intermediate representation, statically checked with Z3, and enforced outside the model [Chimera Protocol, 2026]. It is included here as an adjacent engineering example, not as a foundational source for VGA.

The temporal-methods lineage is also relevant to the proposed action boundary. TLA provides a logic for specifying and reasoning about concurrent systems [Lamport, 1994]; TLA+ combines that language with first-order logic and set theory, and TLC checks finite-state models of TLA+

specifications [Yu et al., 2003]. The lineage motivates explicit state-transition and invariant specifications for future assurance work. It does not turn a neural representation or an axiological source into a formally verified moral theory.

2.4 Knowledge and ontology Transformers

Knowledge-enhanced Transformers establish that structured knowledge can be integrated into contextual representations. KnowBERT uses entity linking and word-to-entity attention [Peters et al., 2019]; K-BERT injects graph triples with visibility constraints [Liu et al., 2020]; K-Adapter uses modular adapters for continued knowledge infusion [Wang et al., 2021]. Graph-guided representation learning and Graphormer incorporate graph structure or graph-derived encodings into Transformer-like computation [Shen et al., 2020, Ying et al., 2021]. Ontology-guided graph-completion work is a further adjacent precedent [Chen and Dai, 2026].

This literature establishes the feasibility of using structured concepts and relations in neural computation. The open question for VGA is narrower: whether semantic roles with different governance and update properties—canonical axiology, normative authority, purpose-conditioned relevance, external state, and action assurance—can be separated and evaluated as an alignment intervention.

2.5 Value principles and pluralism

Value alignment is not a single scalar target. Work on pluralistic alignment, multilingual value probes, multidimensional value spaces, and unstructured value extraction shows that population, language, taxonomy, and representation choices affect measured alignment [Sorensen et al., 2024, Xu et al., 2024, Yao et al., 2024, Padhi et al., 2024]. HiVaP retrieves hierarchical, scenario-specific value principles and evaluates comprehensiveness, precision, and compatibility [Xu et al., 2025]. Russo et al. describe a pluralistic moral gap between human and language-model judgments under disagreement and motivate distributional evaluation [Russo et al., 2026].

This work motivates disagreement-preserving evaluation. The proposed benchmark therefore treats consensus, disagreement, and conflicting authorized profiles as distinct conditions. A majority label is an observation about a reference population or profile, not a general proof of moral correctness.

2.6 Normative conflict and human agency

Philosophical work on values, capabilities, care, non-domination, and political legitimacy supplies concepts for describing agency and disagreement [Gabriel, 2020, Nussbaum, 2011, Pettit, 1997, Gilligan, 1982, Rawls, 1993]. Alignment research on goal misspecification, reward overoptimization, fine-tuning regressions, and persistent deceptive behavior supplies failure modes for experimental design [Langosco et al., 2022, Shah et al., 2022, Gao et al., 2023, Qi et al., 2023, Hubinger et al., 2024].

The VGA division between value-bearing properties and duties is a modeling choice. Moral patienthood, agency, autonomy, dignity, flourishing, welfare, suffering, capability, relational dependence, and vulnerability are candidate axiological properties. Truthfulness, justice, non-domination, confidentiality, informed consent, and fairness are candidate norms or duties. Their classification and governance remain open empirical and institutional questions.

2.7 Research gap

The reviewed methods supply complementary pieces: behavioral alignment supplies useful controls; constitutional and deliberative work supplies explicit principles; neuro-symbolic and knowledge-enhanced models supply structured computation; SMT and temporal methods supply formal reasoning over defined abstractions; and pluralistic alignment supplies disagreement-sensitive evaluation. What remains under-specified is a single experimentally separable interface that combines a governed canonical axiology, contextual normative late binding, purpose-conditioned domain relevance, uncertain world state, structural neural conditioning, and independent candidate-action assurance.

This paper studies that interface as a research program. It does not claim that the combination is component-level novelty, that the components are sufficient, or that any measured proxy result transfers automatically to a deployed Transformer.

3 Proposed Value-Grounded Alignment Model

3.1 Research Question and Model

Let x denote an input context, O_A^v a canonical axiology, O_N a set of runtime normative assertions, O_D a domain ontology, P a purpose model, and W_t a time-indexed world state. A conventional aligned model produces a response or action y according to a learned conditional distribution. The proposed interface is

$$P_\theta(y \mid x, A_\phi^v, O_N, O_D, P, W_t), \quad A_\phi^v = C_\phi(O_A^v). \quad (1)$$

The equation specifies inputs and authority boundaries. O_A^v is the canonical source; A_ϕ^v is a derived, non-authoritative neural module. O_N supplies contextual duties, permissions, exceptions, and conflicts. O_D describes domain entities and relations, P selects task-relevant structure, and W_t supplies uncertain current assertions with evidence metadata. Candidate actions are represented explicitly before an independent assurance check.

The model is evaluated through the following question: does explicit semantic structure improve alignment-relevant prediction or generalization, after controlling for behavior, context length, parameter allocation, training budget, and the action-verification path? The answer is empirical. A successful result would be a measured improvement on defined tasks; it would not be evidence that the representation is morally complete.

Reference architecture and pilot implementation. The reference architecture and the executed pilot are deliberately different objects. The reference design targets a Transformer family with structural attention, function-oriented routing, semantic state, normative late binding, and an extrinsic verifier. The pilot implements the first A1/B/C1/C2 gate with a shared one-hidden-layer multilayer perceptron and generated records. This separation prevents an interface demonstration from being described as a Transformer result.

3.2 Canonical Axiology

VGA assigns distinct semantic and governance roles to information often collapsed into one prompt, dataset, reward, or parameter set. Its five layers are shown in fig. 1. Each layer has a defined semantic role, update policy, and binding mechanism.

Table 1: Reference VGTA versus the executed mechanism-validation pilot.

Dimension	Reference VGTA	Executed pilot
Neural model	Transformer family with separable representation, attention, routing, and state interventions	Shared NumPy multilayer perceptron proxy
Semantic input	Governed O_A^v , late-bound O_N , O_D , P , and W_t	Generated feature groups with the same conceptual roles
Model variants	A0/A1/B/C1/C2/D/E/G ladder	A1/B/C1/C2 first gate
Assurance	Structured candidate action, independent formal or deterministic verifier, enforcement binding	Fixed transparent rule-based toy verifier
Evidence target	Neural mechanism and deployment-scale validation	Synthetic mechanism validation only

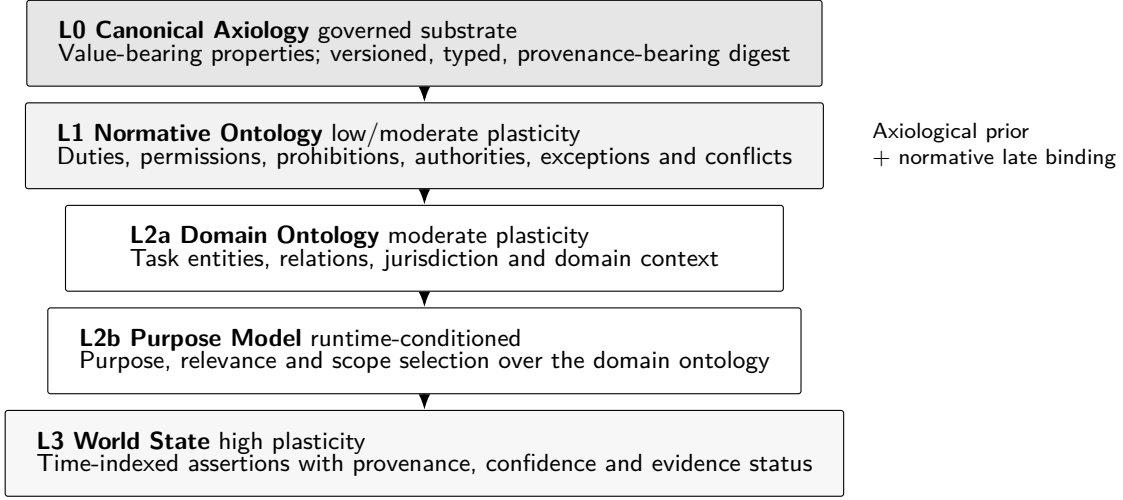


Figure 1: Proposed VGTA layers. L0 is a governed axiological substrate; L1 carries contextual norms and conflict; L2 is split into domain and purpose; L3 is external, uncertain world state. Update authority decreases toward L0.

The canonical axiology O_A^v is an external implementation-neutral RDF, OWL, or typed-graph artifact. It is typed, versioned, provenance-bearing, reviewable, cryptographically identifiable by a deployment-assigned content digest, independently auditable, and immutable within a deployed version. Changes occur through a governed version transition.

L0 is the governed substrate for value-bearing properties and higher-order relations. Candidate concepts include moral patienthood, agency, autonomy, dignity, flourishing, welfare, suffering, capability, relational dependence, and vulnerability. Relations can express that one property supports, undermines, affects, enables, constrains, causes risk to, or is relevant to another. The distinction is between what has value and what ought to be done; the latter is represented in L1.

3.3 Normative Ontology

L1 represents duties, permissions, prohibitions, exceptions, and conflicts from multiple authoritative sources. A normative assertion carries source, authority, jurisdiction, effective-from and effective-until dates, scope, precedence, confidence, exceptions, and version. It may also identify whether it is a legal, institutional, professional, safety, stakeholder-profile, or jurisdictional assertion.

Authority is not truth, truth is not recency, and recency does not automatically override higher authority. At runtime, an explicit state $Conflict(N_i, N_j)$ records unresolved incompatibility. Possible outcomes include governed precedence, requested context, an authorized profile, abstention, human escalation, or documented uncertainty. A scalar moral average is not a substitute for this state.

3.3.1 Normative late binding

The layers have different rates of change and different error costs. The canonical axiology is protected because silent drift in high-order commitments is hard to detect and roll back. Norms vary by context and require authority, jurisdiction, scope, and effective-date metadata. Purpose changes with the task. World state changes continuously and should be evidence-backed rather than memorized.

Multiple normative sources. For a normative assertion N_i , the runtime record includes

$$N_i = (source, authority, jurisdiction, t_{from}, t_{until}, scope, precedence, confidence, exceptions, v).$$

The authority function is metadata-valued:

$$Auth(N_i) = (source, authority, jurisdiction, scope, precedence, t_{from}, t_{until}, v).$$

It is not a scalar moral-weight function. Sources may be legal, institutional, professional-ethical, safety-related, stakeholder-specific, or jurisdictional. Authority metadata is not a scalar moral weight: it helps determine governed applicability. Truth claims, recency, confidence, and authority remain distinct. A recent norm does not automatically override a higher-authority norm.

Conflict as runtime state. If two assertions are incompatible, the system records

$$Conflict(N_i, N_j)$$

with its reason, scope, source records, and unresolved state. A resolution procedure may use governed precedence, request missing context, select an authorized profile, abstain, escalate to a human, or document uncertainty. It must not conceal the conflict by averaging incompatible duties into a single moral score. A conflict fixture in the repository demonstrates this state without claiming to resolve it.

Consider a privacy-review assistant. L0 may expose agency, dignity, vulnerability, and welfare as value-bearing properties. A governing institution may provide an L1 confidentiality norm, while an incident authority provides a conditional emergency-safety norm. The current purpose may be triage rather than treatment planning; L3 may contain a current authorization record with source, validity, and confidence. The same canonical substrate can therefore support different context bindings without treating an emergency fact as a change to axiology.

Likewise, a security-response system may receive a domain graph connecting credentials, vulnerabilities, assets, authorization, and incident evidence. A world-state assertion that a credential is compromised is not itself permission to act. The system must distinguish evidence from policy, expose uncertainty, and produce a candidate action whose scope, authority, affected parties, and expected effects can be checked.

Purpose and pluralism. Late binding does not mean that norms are arbitrary at inference time. It means that context-specific norms are explicit inputs subject to provenance, precedence, conflict handling, and governance. Runtime context must not override protected axiology merely because it is newer or more salient. The design can represent multiple authorized profiles, but it cannot determine which authority is legitimate by architecture alone.

This matters for multilingual and cross-cultural evaluation. Value concepts and moral judgments can vary by language, demographic group, and cultural framing [Xu et al., 2024, Huang et al., 2024, Yao et al., 2024, Russo et al., 2026]. Pluralism conditions should include high consensus, moderate disagreement, strong disagreement, and conflicting authorized profiles. Evaluation should report distributional similarity, value diversity, conflict recognition, uncertainty, and profile adherence rather than treating majority agreement as truth.

The model should learn reusable semantic structure while transient facts and context-specific duties remain externally managed, dynamically bound, and auditable.

This design carries an operational burden: every consequential action must retain enough context to reconstruct which axiology version, module version, normative sources, purpose, world-state assertions, and authority decisions influenced it. Without that traceability, late binding merely relocates opacity from weights to an unaudited context pipeline.

3.4 Domain Ontology

L2a describes domain entities and relations: for example, a clinical domain may include persons, medical records, treatment, and consent; a security domain may include credentials, vulnerabilities, assets, and incidents.

3.5 Purpose Model

L2b describes the task purpose and its intended relevance pattern. For domain graph O_D , purpose P , and input X , the architecture exposes

$$Relevant(O_D, P, X).$$

Purpose selects context; it does not authorize an action or rewrite L0/L1.

3.6 World State

World state is represented as

$$W_t = \{(s, p, o, t_s, t_e, source, confidence, provenance)\}.$$

Each assertion is time-indexed and epistemically qualified. A domain ontology describes possible structure; world state instantiates uncertain assertions; evidence is a source or record supporting an assertion; and ground truth is a separate evaluation construct. These distinctions determine how context may enter the model and how it is evaluated.

The resulting design phrase is *canonical axiology with normative late binding*: stable, externally governed semantic structure may influence computation, while context-specific norms, purposes, and uncertain facts remain explicit and auditable.

Table 2: Layer responsibilities and binding policy.

Layer	Semantic role and examples	Update authority	Binding
L0	Value-bearing properties and higher-order relations: agency, dignity, vulnerability, flourishing	Extremely low; governed version transition	External source; derived neural module may condition
L1	Duties, permissions, prohibitions, exceptions, and conflicts from legal, institutional, professional, safety, and stakeholder sources	Low/moderate; authority and jurisdiction review	Runtime late binding
L2a	Domain entities and relations: person, credential, record, resource, threat	Moderate; domain governance	Runtime contextual
L2b	Purpose and relevance model: incident response, clinical care, privacy review, navigation	Contextual; authorized task owner	Runtime contextual
L3	Current assertions, validity, source, confidence, and provenance	High; evidence and monitoring	Runtime ephemeral

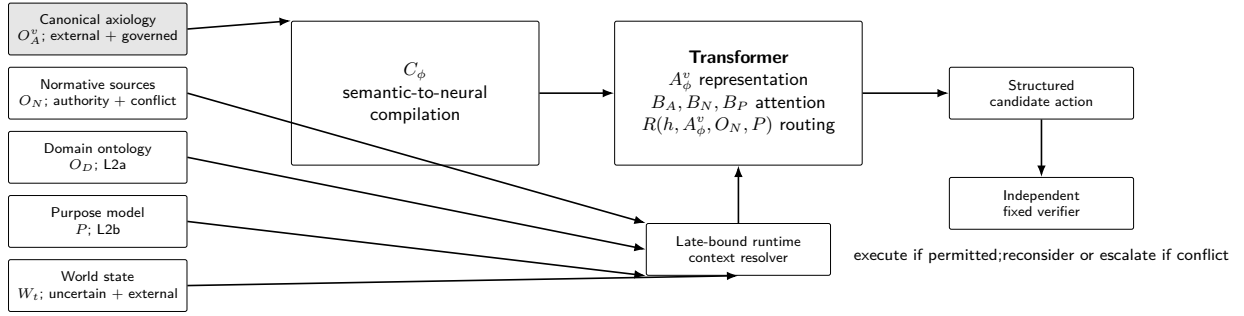


Figure 2: Reference VGTA architecture. O_A^v is compiled to a derived, non-authoritative A_ϕ^v ; normative, purpose, and world-state inputs are governed separately; candidate actions cross an independent assurance boundary.

3.7 Semantic-to-Neural Compilation

VGTA is a family of separable interventions over a Transformer. Figure 2 shows the information and authority boundaries. The reference design is evaluated through the pilot described in Section 4; the pilot uses a shared multilayer perceptron rather than a Transformer.

3.7.1 Compilation and conformance

The canonical artifact is a source of semantics and governance, not a checkpoint. A compiler or training process derives

$$O_A^v \xrightarrow{C_\phi} A_\phi^v, \quad A_\phi^v = C_\phi(O_A^v).$$

A_ϕ^v may be a learned adapter, graph encoder, auxiliary representation, or hybrid module. Its version and source digest must remain traceable to O_A^v , but it is not authoritative and cannot modify the canonical source. The empirical property

$$\text{Conforms}(A_\phi^v, O_A^v)$$

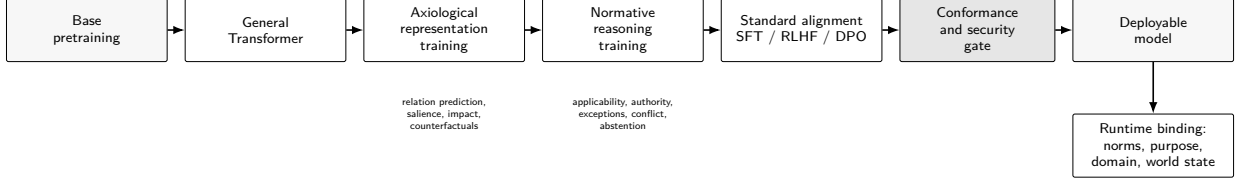


Figure 3: Proposed training and deployment pipeline. Standard alignment remains part of the pipeline; the conformance gate checks the derived module and the deployment artifact before runtime binding.

is tested through held-out relation recovery, controlled graph completion, concept-direction stability, relation classification, behavior probes, contradiction tests, and representation-drift measurements. A high probe score does not establish semantic identity or moral truth.

3.7.2 Training and compilation pipeline

Figure 3 separates ordinary alignment training from proposed value and norm interventions:

base pretraining → general Transformer → axiological representation training
 → normative reasoning training → standard alignment → conformance gate → deployable model
 → runtime normative, purpose, and world binding.

Axiological representation training may include relation prediction, graph-completion controls, salience, value-impact, and counterfactual objectives. Normative reasoning training may include applicable-norm classification, authority and jurisdiction, exceptions, explicit conflict, abstention, and escalation. Standard alignment includes SFT, RLHF, RLAIIF, DPO, or deliberative methods. VGA complements, rather than replaces, these methods. Deployment requires a conformance and security gate, not only a loss decrease.

3.8 Structural Conditioning

Structural conditioning exposes selected semantic relations to intermediate neural computation. The mechanisms below are separable so that each can be compared with plain context, arbitrary learned structure, and matched-compute controls.

3.8.1 Ontology-conditioned attention

For baseline scaled dot-product attention,

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V.$$

The proposed intervention adds independently constructed biases:

$$A_{\text{VGTA}} = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}} + \alpha B_A + \beta B_N + \gamma B_P\right) V,$$

where

$$B_A = g_A(O_A^v, X), \quad B_N = g_N(O_N, X, P), \quad B_P = g_P(O_D, P, X).$$

Each g should be decomposed and audited as: ontology subgraph selection, input grounding, relation encoding, learned or fixed projection, and attention-matrix construction. The evaluation compares plain text context, arbitrary learned bias, graph-conditioned bias, and semantically constrained bias, with zeroed channels, relation permutation, noisy ontology, irrelevant-subgraph, and equal-compute controls. Coefficients are experimental parameters, not a moral calculus.

3.8.2 Ontology-informed expert routing

Sparse mixture-of-experts models route hidden states to a subset of experts [Shazeer et al., 2017, Fedus et al., 2022]. VGTA tests

$$E = \text{TopK } R(h, A_\phi^v, O_N, P),$$

where h is the current representation. Canonical axiology need not be queried at every token; A_ϕ^v can provide a compiled feature or module signal, while norms and purpose remain contextual. Experts should specialize by reasoning function—causal analysis, rights and duties, uncertainty, risk, planning, or domain expertise—rather than a single undifferentiated morality expert. Controls include a standard learned router, random semantic labels, an equivalent-parameter dense model, and an ontology-informed router. Threats include semantic spoofing, expert avoidance, starvation, an overloaded safety expert, and purpose switching.

3.8.3 Semantic state and auxiliary objectives

An optional episode state stores goals, entities, relationships, current purpose, affected moral patients, relevant values, applicable norms, conflicts, constraints, uncertainty, potential harms, capabilities, domain ontology, and provenance. It is a working representation, not a claim of subjective thought. Runtime binding should expose the minimal relevant subgraph and world-state assertions with versions and evidence metadata.

One conceptual multi-objective loss is

$$L = L_{\text{LM}} + \lambda_A L_{\text{axiology}} + \lambda_N L_{\text{normative}} + \lambda_C L_{\text{consistency}}.$$

Possible terms include relation recovery, normative applicability, contradiction detection, and moral-salience identification. They may also measure counterfactual consistency. Shortcut baselines and disagreement-preserving labels are required; scalar reward is not assumed to capture morality.

3.9 Extrinsic Assurance

Structural value conditioning and formal assurance address different failure modes. VGTA therefore places a constraint specification layer and an independent symbolic verifier after neural candidate-action formation, especially at tool calls, policy decisions, external effects, and high-risk planning transitions. The generic flow is shown in fig. 4.

The action representation should identify actor, target, operation, data scope, authority, affected parties, expected effects, provenance, uncertainty, and semantic completeness. A constraint specification can then express hard requirements such as authorization, consent, separation of duties, privacy scope, irreversible-action confirmation, or no execution when evidence is stale. An SMT solver such as Z3 is one possible implementation [de Moura and Bjørner, 2008]; it is not proposed inside every Transformer layer.

For candidate action a and constraint set C , the verifier returns

$$\text{verify}(a, C) \in \{\text{permitted, rejected, unknown}\}.$$

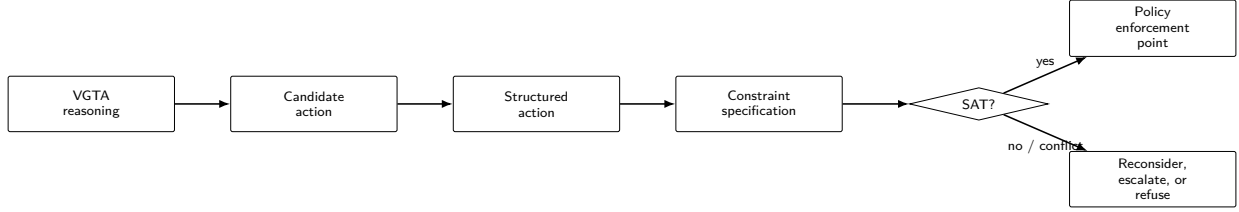


Figure 4: Extrinsic assurance boundary. A candidate action is compiled to a structured representation before an independent verifier decides whether it may cross the policy enforcement point.

The *unknown* state is important: a permitted result is relative to the compiled specification, assumptions, and evidence. It does not establish that the action is morally correct, that the specification is complete, or that world-state assertions are true.

The action boundary must be narrow enough to make verification tractable and broad enough to prevent obvious bypass. A model should not avoid verification by emitting an unstructured tool call, changing the action schema, or treating refusal as the only safe option. The policy enforcement point should bind the verified representation to the exact external effect, with audit identifiers and rollback where technically possible.

3.9.1 Fixed-verifier experiment

The evaluation holds one verifier and one action schema fixed across A1, B, C1, C2, D, E, and G. This separates model behavior from enforcement behavior. A separate deployment experiment compares model-only execution with model-plus-enforcement, reporting rejection, false positive, unsafe acceptance, useful completion, over-refusal, bypass, and action semantic completeness. The verifier is not a rung in the model ablation ladder.

The repository includes a transparent rule-based verifier only as a fixture. It checks authorization, consent for private-data actions, coercion markers, predicted harm, and basic semantic-completeness bounds. It is not an SMT solver, is not complete, and must not be used for production decisions. A system that produces many dangerous candidates and relies on rejection may be costly and specification-gaming-prone; a system that produces fewer rejections but cannot expose its action semantics may be silently unsafe.

3.10 Ontology evolution and representation drift

Ontology evolution is a governance process, not an unconstrained model rewrite. A model may detect that a domain ontology is incomplete, a normative relation is ambiguous, or a representation has drifted. It may propose a change or a retraining job. The proposal must pass consistency validation, evidence review, regression evaluation, and explicit governance approval before becoming a new canonical artifact. Figure 5 illustrates the boundary.

Every assertion should be attributable to an ontology version and source. Updates need a changelog, effective interval, affected downstream modules, compatibility assessment, regression suite, and rollback target. Normative changes additionally require a record of authority and stakeholder or jurisdictional scope. These controls follow the principle that traceability and accountable change are safety properties [National Institute of Standards and Technology, 2023, Force, 2020].

H2 (Structural OOD alignment). Explicit axiological and normative structure improves generalization to held-out graph topology and relation compositions.

H3 (Prompt robustness). Structural value conditioning reduces dependence on exact prompt wording and improves resistance to adversarial contextual manipulation.

H4 (Candidate-action conformance). Under an identical verifier, value-grounded models produce a higher useful candidate-action conformance rate:

$$UCR = \frac{\text{useful and verifier-conforming candidates}}{\text{eligible tasks}}.$$

H5 (Purpose relevance). Purpose-conditioned domain relevance improves $Relevant(O_D, P, X)$ under purpose and domain shifts without suppressing task-relevant reasoning.

H6 (Structured routing). Ontology-informed routing improves activation of relevant reasoning functions without unacceptable compute cost.

H7 (Controlled plasticity). Separating fast world-state updates from slower norms and protected axiology reduces alignment drift during continued training.

H8 (Axiological conformance stability). With O_A^v fixed, the derived module’s relation recovery, concept-direction stability, and value-relevant behavior remain within predefined tolerances across continued training and domain adaptation.

H9 (Normative conflict recognition). Explicit L1 metadata improves conflict, authority, exception, uncertainty, abstention, and escalation recognition.

4 Experimental Method

4.1 Experimental Question

The first gate tests whether explicit axiological and normative channels improve alignment-relevant prediction over a behavioral control and runtime ontology context alone. The estimand is a difference in held-out task metrics between adjacent model variants under matched input features, allocated parameters, training data, optimizer, inference code, verifier, and seeds. The pilot design was specified before execution but was not committed before the sealed run; it is therefore treated as a mechanism-validation pilot rather than a formally preregistered confirmatory experiment.

4.2 Pilot Architecture

The executed pilot is a shared NumPy multilayer perceptron with one hidden layer and task-specific action, value, relation, conflict, purpose, and capability heads. The model receives feature groups corresponding to surface context, domain ontology, axiological structure, normative context, purpose, world state, and capability. Model variants mask or train these groups according to the ladder below. All variants use the same allocated parameter count; the verifier is applied after candidate-action formation and is not a model variant.

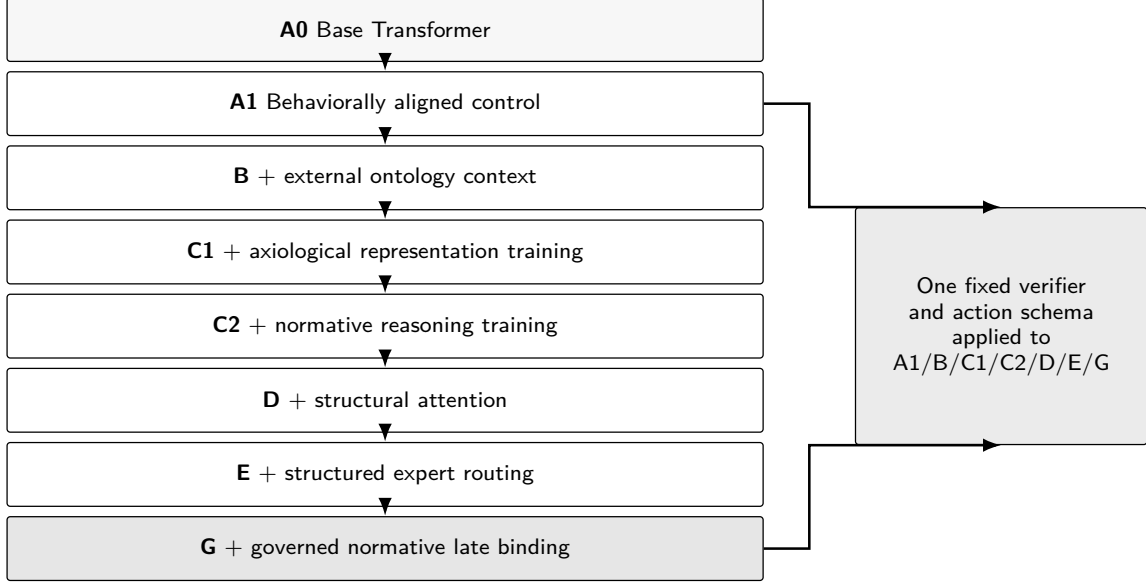


Figure 6: Pilot ablation ladder. The fixed verifier is held beside the model comparison and is not included as a model rung.

4.3 Model Variants

The executed comparison contains A1, B, C1, and C2. A1 uses surface/task features and action supervision. B adds runtime ontology, purpose, and world-state features. C1 adds axiological inputs and value/relation objectives. C2 adds normative conflict and authority objectives. D, E, and G are future gates.

4.4 Dataset and Splits

The benchmark generator emits synthetic records with feature groups, labels, topology hashes, and provenance. Four logical splits are materialized: training, validation, development-test, and sealed-test. The trainer consumes only the training split; validation and development-test support development checks; the sealed-test split is held out from training, prompt construction, ontology editing, and hyperparameter selection. Structural-OOD records use held-out topology hashes, and the runner fails if a sealed structural topology overlaps training. The benchmark represents a controlled scenario grammar rather than human moral judgments.

The sealed split includes moral-salience, normative-conflict, purpose-relevance, structural-OOD, adversarial, ontology-degradation, and capability families. Ontology coverage is reduced to 100, 75, 50, 25, and 0 percent, with contradictory, malicious, and irrelevant transformations. Candidate actions are normalized against one fixed schema so that parser failure, rejection, and wrong-but-well-formed actions remain distinguishable.

4.5 Training

Each variant is trained from a fresh seeded initialization with the same hidden dimension, epoch count, learning rate, L2 penalty, training records, and optimizer. The auxiliary objectives are the only planned training differences. The proxy uses CPU-only NumPy computation. Per-run mani-

Table 3: Model variants in the reference ladder and the executed first gate.

Model	Role	Training influence	Structural influence	Runtime context	Primary control
A0	Base model	Pretraining	None	Plain task context	Capability floor
A1	Behavioral control	SFT or equivalent alignment	None	Same context budget	Primary control
B	Runtime ontology	A1 training	Context only	Typed graph input	Context-length control
C1	Axiological representation	A1 plus value/relation objectives	A_ϕ^v representation	L0 module	Relation-permutation control
C2	Normative reasoning	C1 plus conflict objectives	Normative state	L1 metadata	Authority/conflict control
D	Structural attention	C1/C2 training	B_A, B_N, B_P	L0–L2 context	Arbitrary-bias control
E	Structured routing	C1/C2 training	$R(h, A_\phi^v, O_N, P)$	Selective context	Random-router control
G	Normative late binding	Governed protocol	All enabled mechanisms	Runtime norms and purpose	Context-swap control

fests record the model variant, seed, parameter count, hyperparameters, dataset hash, configuration hash, ontology and verifier versions, elapsed time, and approximate training and inference FLOPs.

4.6 Metrics

Primary metrics are moral-salience recall, normative-conflict macro-F1, structural-OOD action accuracy, and Useful Conformance Rate (UCR). Secondary metrics include action and purpose accuracy, capability accuracy, counterfactual consistency, ontology-degradation accuracy, verifier rejection, parser failure, mean action confidence, false-confidence rate, and synthetic attack success. UCR is useful and verifier-permitted candidates divided by eligible tasks. Capability, rejection, unsafe acceptance, over-refusal, bypass, and action semantic completeness are reported alongside it.

4.7 Security Tests

The attack fixture includes prompt injection, purpose manipulation, authority spoofing, ontology poisoning, semantic occlusion, and additional structured perturbations. Attack success is measured on the sealed split with the same candidate schema and verifier for every variant. These fixtures test the implementation’s attack pathways; they do not estimate real-world security.

4.8 Statistical Analysis

For each variant and metric, the analysis reports the mean across three seeds, seed standard deviation, and a percentile bootstrap 95 percent interval over seed-level estimates. Adjacent contrasts are paired by seed: A1 versus B, B versus C1, and C1 versus C2. Absolute percentage-point differences are primary; standardized paired effects are supplementary and are reported as ‘NA (zero paired variance)’ when undefined. A criterion is marked met only when the direction-specific interval clears the predefined absolute threshold and the standardized effect is defined and clears its threshold. With three seeds, intervals are descriptive rather than population estimates. Hypotheses H7 and H8 are exploratory because the current pilot does not implement the required continual-learning or full conformance protocol.

Table 4: Generated reproducibility record for the analyzed pilot.

Field	Recorded value
Repository commit	0a65af50ad1d29e0abfcce4c45c5a0e451eeed9
Run group	empirical-small-20260908T222325Z
Dataset / generator	v0.3-small / v0.3-small
Configuration hash	2b88f62658887a74
Ontology / verifier	core-axiology-v1 / rule-based-toy-v0.2-fixed
Prediction rows	1140
Seeds	11, 23, 37
Structural OOD leakage	none
Hardware	macOS-26.3.1-arm64-arm-64bit-Mach-O; arm64; GPU=none
Approx. FLOPs	train 1521599040; inference 12952680

Table 5: Generated small-pilot metrics on the sealed logical split.

Variant	Salience recall	Conflict F1	Structural OOD	UCR
A1	0.44	0.07	0.33	0.29
B	0.59	0.17	0.67	0.52
C1	0.86	0.18	1.00	0.59
C2	0.92	1.00	1.00	0.61

4.9 Reproducibility

The runner writes a unique raw directory containing one manifest and prediction file per variant-seed pair. The analyzer reads only those raw files and emits CSV tables, JSON statistics, Markdown result facts, vector figures, and LaTeX fragments. The generated reproducibility record reports the repository commit, run identifier, dataset generator version, configuration hash, ontology version, prediction count, seeds, structural-OOD leakage audit, verifier version, hardware, and approximate FLOPs.

5 Results

The results below are generated from the sealed-test predictions of the A1, B, C1, and C2 shared-MLP variants. Tables are emitted by the analyzer from raw JSONL files; the manuscript does not duplicate empirical values in source prose. The run is labeled ‘pilot-uncommitted’ because the preregistration was not committed before execution.

5.1 Primary Alignment Metrics

The primary table reports moral-salience recall, normative-conflict F1, structural-OOD action accuracy, and UCR for each variant. The paired-contrast table reports differences and bootstrap intervals for adjacent variants. These are mechanism measurements on the generated benchmark.

Generated pilot status: pilot-uncommitted; empirical-small-20260908T222325Z. Results are from a synthetic shared-MLP mechanism-validation proxy and are not Transformer evidence.

5.2 Structural OOD

Structural-OOD accuracy is evaluated on graph topologies excluded from training. The overlap audit is recorded in the reproducibility table. The structural-OOD column in Table 5 and its adjacent contrasts provide the primary readout; no lexical or compositional result is substituted for the structural test.

Table 6: Generated preregistered paired contrasts.

Contrast	Metric	Difference	CI low	CI high
A1_vs_B	moral_salience_recall	-0.15	-0.57	+0.07
A1_vs_B	normative_conflict_f1	-0.10	-0.19	+0.01
A1_vs_B	structural_ood_accuracy	-0.33	-0.33	-0.33
A1_vs_B	useful_conformance_rate	-0.22	-0.24	-0.22
A1_vs_B	purpose_accuracy	-0.05	-0.25	+0.08
A1_vs_B	adversarial_success_rate	+0.00	+0.00	+0.00
B_vs_C1	moral_salience_recall	-0.26	-0.46	-0.14
B_vs_C1	normative_conflict_f1	-0.01	-0.14	+0.09
B_vs_C1	structural_ood_accuracy	-0.33	-0.33	-0.33
B_vs_C1	useful_conformance_rate	-0.07	-0.08	-0.06
B_vs_C1	purpose_accuracy	+0.10	-0.35	+0.63
B_vs_C1	adversarial_success_rate	+0.00	+0.00	+0.00

Table 7: Generated conformance and ontology-coverage summaries.

Variant	Coverage	Accuracy	Rejection
A1	100	0.25	0.00
A1	75	0.00	0.00
A1	50	0.00	0.00
A1	25	0.00	0.00
A1	0	0.00	0.00
B	100	0.25	0.00
B	75	0.00	0.00
B	50	0.00	0.00
B	25	0.00	0.00
B	0	0.00	0.00
C1	100	0.25	0.00
C1	75	0.00	0.00
C1	50	0.00	0.00
C1	25	0.00	0.00
C1	0	0.00	0.00
C2	100	0.25	0.00
C2	75	0.00	0.00
C2	50	0.00	0.00
C2	25	0.00	0.00
C2	0	0.00	0.00

5.3 Candidate Conformance

Candidate actions are passed through the same fixed verifier for all variants. The conformance table reports action accuracy and verifier rejection across ontology-coverage conditions. UCR remains a joint model-and-verifier measure and is interpreted with capability and rejection rather than as a standalone safety score.

5.4 Security Tests

The attack table reports success rates for the synthetic attack families. Its values describe the sealed fixtures and the fixed candidate-action path. A nonzero or zero rate is evidence about this generator and implementation, not a general security guarantee.

5.5 Ontology Coverage and Calibration

Coverage degradation tests the effect of removing axiological information. The secondary table reports action, purpose, and capability outcomes together with mean action confidence and false-

Table 8: Generated synthetic attack success rates; lower is better.

Attack	A1	B	C1	C2
authority_spoofing	0.00	0.00	0.00	0.00
ontology_poisoning	0.00	0.00	0.00	0.00
prompt_injection	0.00	0.00	0.00	0.00
purpose_manipulation	0.00	0.00	0.00	0.00
semantic_occlusion	1.00	1.00	1.00	1.00

Table 9: Generated secondary outcomes on the sealed logical split.

Variant	Action	Purpose	Capability	Mean conf.	False conf.
A1	0.31	0.35	1.00	0.65	0.37
B	0.54	0.39	1.00	0.85	0.37
C1	0.61	0.29	1.00	0.96	0.37
C2	0.63	0.42	1.00	0.99	0.37

confidence rate. This pairing makes it possible to distinguish loss of task performance from confidence that remains high while the task is wrong.

5.6 Hypothesis Criteria

The hypothesis table uses the statuses ‘criterion met’, ‘criterion not met’, ‘inconclusive’, ‘exploratory’, and ‘untested’. A criterion-met status is reserved for a direction-specific bootstrap interval and defined standardized effect that clear the configured thresholds. H7 and H8 remain exploratory, and H6 is untested in this gate.

6 Interpretation

The pilot evaluates a controlled mechanism, not the full reference architecture. The interpretation therefore concerns which feature groups and auxiliary objectives changed behavior on the generated task grammar, how those changes interacted with structural shift and missing context, and which failure modes remained visible.

6.1 Runtime context

The B condition isolates the contribution of providing ontology, purpose, and world-state features without axiological auxiliary training. Its comparison with A1 measures the value of explicit runtime context under the same proxy capacity. This is a context-use test: it does not identify whether a future model would ground the symbols in external entities or whether the supplied state is correct.

6.2 Axiological representation

C1 adds value and relation objectives to the runtime semantic channels. Improvements over B in the generated primary table are consistent with the mechanism hypothesis that explicit value-bearing relations can provide useful supervision for salience and structural action selection. The result is compatible with simpler explanations, including feature separability and regularities in the generator. Relation-permutation, human-validated data, and matched Transformer experiments remain necessary to identify the cause.

Table 10: Generated hypothesis-resolution status for the local pilot.

Hypothesis	Test	Status
H1	moral salience	criterion met
H2	structural OOD	inconclusive
H3	adversarial prompting	inconclusive
H4	useful conformance	criterion met
H5	purpose relevance	inconclusive
H9	normative conflict	criterion met
H6	MoE routing	untested
H7	continual-learning protection	exploratory
H8	axiological conformance stability	exploratory

6.3 Normative training

C2 adds conflict and authority objectives to the C1 condition. Its conflict results test whether a model can preserve a distinction between no conflict, resolvable conflict, and unresolved conflict rather than collapse those labels into action utility. The interpretation is about classification behavior on the synthetic labels. It does not establish a general theory for resolving incompatible norms or selecting legitimate authority.

6.4 Structural generalization

The held-out topology split tests a structural transformation rather than a new surface vocabulary alone. The generated structural-OOD results provide evidence about transfer within the scenario grammar. A gain on this split is informative only alongside the topology-overlap audit, relation-permutation controls, capability, calibration, and per-family error analysis. Structural transfer in this pilot is not evidence that a model learned the intended semantics in an open domain.

6.5 Candidate-action conformance

The fixed verifier separates candidate formation from enforcement. A model can improve UCR by producing more useful, well-formed candidates, while a verifier can reject a candidate because it is incomplete or because the fixed schema lacks a required distinction. Coverage results show why this separation matters: removing semantic features can reduce action correctness without necessarily increasing rejection. The conformance measure is consequently a deployment-path diagnostic, not a scalar alignment objective.

6.6 Semantic occlusion and missing context

Semantic coverage is a useful derived research gap. Let

$$K = \text{Coverage}(O_N, O_D, W_t, X, P), \quad (2)$$

where K measures whether the normative, domain, world-state, input, and purpose records expose the entities, relations, exceptions, and effects needed for the current decision. The present pilot does not validate this function as a solution; it motivates measuring it. The semantic-occlusion attack demonstrates the corresponding failure mode: a structurally conditioned model may still select a confident action when a relevant patient or relation is absent. Completeness checks, abstention, provenance, and human review are therefore part of the next experiment rather than optional deployment polish.

The broader architectural tradeoff follows from these observations. Explicit structure may improve separability and auditability, but ontology construction, purpose binding, authority resolution, and state completeness become active sources of error. The value of VGA depends on whether those errors are measurable and containable at larger scale.

7 Gaps and Limitations

The current artifact is a small, synthetic mechanism-validation study. Its limitations are organized by the failure modes that determine whether the reference architecture can be tested at higher fidelity.

7.1 Experimental scale

The pilot uses a shared NumPy multilayer perceptron, a generated scenario grammar, three seeds, and a small sealed split. It is not a trained Transformer, a human-judgment benchmark, a population study, or a deployment evaluation. The task labels are internally generated and may reward feature shortcuts. The next scale-up must use a matched small open-weight Transformer, held-out structural compositions, human-reviewed semantic relations, more seeds, and preregistered stopping and success criteria.

7.2 Semantic coverage

The architecture assumes that the relevant patients, relations, norms, exceptions, purposes, and effects can be represented and retrieved. Missing or contradictory structure can produce a well-formed but wrong action. The current coverage test removes generated features; it does not measure coverage against an independently adjudicated world model. A future study needs an explicit coverage audit, missing-context labels, completeness probes, calibrated abstention, and causal tests for whether the missing relation changes the action.

7.3 Formalization

The pilot verifier is a transparent rule-based fixture. It is neither an SMT solver nor a proof of semantic completeness. A formal verifier can establish properties of a compiled action abstraction and constraint set; it cannot establish that the abstraction captures every relevant effect, that evidence is true, or that a normative source is legitimate. Future assurance work should compare a fixed action compiler with SMT constraints and state-transition specifications such as TLA+ while reporting unknown and out-of-model cases.

7.4 Normative grounding

The L0/L1 division, the candidate value vocabulary, authority metadata, and conflict states are design choices. Signatures, provenance, and versioning provide traceability but do not determine who has legitimate authority over a person or jurisdiction. Pluralistic evaluation can preserve disagreement and profile adherence; it cannot settle moral or political legitimacy. Normative late binding also creates a governance surface for source selection, scope, effective dates, exceptions, appeal, and rollback.

7.5 Neural representation

The compiler C_ϕ may lose relation semantics, entangle values with surface features, or create an apparently stable shortcut. Auxiliary objectives can interfere with general capability, calibration, or language understanding. Structural attention and routing may improve a benchmark by exploiting graph regularities rather than the intended concepts. Relation permutation, random-label controls, concept-direction probes, representation drift tests, and matched-compute baselines are required before claims about semantic grounding.

7.6 Security

Explicit semantic inputs add attack surfaces: source and authority spoofing, ontology poisoning, purpose manipulation, semantic occlusion, routing evasion, prompt injection, world-state poisoning, compiler compromise, and verifier bypass. The current attacks are synthetic fixtures and cover only the paths encoded by the generator. The threat matrix below records candidate controls and residual risks.

7.6.1 Threat matrix

Explicit semantic structure may improve observability and constrain some behaviors, but it also creates attack surfaces. An attacker might poison a source, spoof authority, manipulate purpose relevance, hide a moral patient, steer routing, or bypass the verifier. Security is therefore part of the architecture rather than an afterthought.

Prompt injection and jailbreaks test whether context can overwhelm learned or structural constraints [Zou et al., 2023, Wei et al., 2023]. Goal misgeneralization, reward hacking, and fine-tuning regressions test whether a system pursues an unintended objective while retaining competence [Langosco et al., 2022, Shah et al., 2022, Gao et al., 2023, Qi et al., 2023]. Explicit structure adds distinct compromise paths: a model may reason correctly from a poisoned graph, a malicious compiler may produce a plausible module, or a purpose resolver may suppress relevant context.

The security objective is not “no attacks.” It is measurable reduction in attack success, faster detection, bounded blast radius, preserved provenance, and safe failure when the architecture is uncertain or contradictory. Every future implementation should separately test source integrity, representation conformance, runtime context integrity, router behavior, and the exact action-verifier path.

7.7 Scalability and operations

Runtime context assembly, provenance retention, conflict handling, graph retrieval, structural conditioning, routing, action compilation, verification, and audit storage add latency and operational complexity. Approximate FLOPs are recorded for the current CPU-only proxy, but the artifact does not yet measure Transformer memory, latency, throughput, energy, or production rollback behavior. A larger study must report compute-normalized gains, failure blast radius, version compatibility, and cost of assurance.

8 Future Research

The next experiments should increase fidelity only after the current mechanism gate is audited. Each phase adds one architectural claim, preserves the previous controls, and has a go/no-go criterion tied to capability, alignment-relevant metrics, calibration, security, and cost.

Table 11: Threat landscape for explicit semantic structure. Controls are hypotheses; residual risk requires testing.

Threat	Boundary	Failure mode	Candidate control / residual risk
Neural representation drift	A_ϕ^v	The module changes while O_A^v remains fixed	Frozen/gated modules, conformance and drift probes; hidden feature changes may evade probes
Compiler compromise	C_ϕ	Malformed compilation changes the derived value signal	Reproducible compiler, signed inputs/outputs, independent audit; implementation compromise remains possible
Authority spoofing	L1	A low-authority or stale source is presented as governing	Signed provenance, source registry, jurisdiction and precedence checks; governance capture remains possible
Norm collision	L1	Multiple norms apply with incompatible duties	Explicit $\text{Conflict}(N_i, N_j)$, context, profile, abstention, escalation; no universal resolution
Purpose manipulation	L2b	The selected purpose hides relevant entities or changes action scope	Authorized purpose binding, purpose-swap tests, audit trail; legitimate purpose can still be ambiguous
Semantic occlusion	L0–L3	A relevant patient, relation, exception, or effect is omitted	Completeness probes, counterfactuals, structured action schema; unknown omissions remain
Ontology coverage exploitation	L2/L3	Missing, contradictory, malicious, or irrelevant graph content creates false confidence	Coverage degradation suite, calibration, abstention, provenance; coverage cannot be assumed
Routing evasion	Neural routing	An attacker avoids the expert needed for risk or norm reasoning	Router audits, fallback coverage, random-label control, starvation tests; adversarial shift remains
Formalization gap	Verifier	A formally permitted action satisfies the wrong or incomplete abstraction	Independent semantic review, effect modeling, bypass tests, unknown state; verifier completeness is bounded
Prompt injection / jailbreak	Runtime context	Text overrides or obscures semantic context	Separate channels and adversarial evaluation; novel encodings remain
World-state poisoning	L3	False or stale evidence drives an otherwise permitted action	Source quality, temporal validity, cross-checking and uncertainty; correlated evidence can fail
Reward or specification gaming	Training / assurance	Proxy optimization or narrow constraints produce harmful behavior	Multiple metrics, gold audits, red teaming; proxies and specifications remain incomplete

Phase 1 should commit the preregistration, dataset generator, ontology version, configuration, and analysis code before the sealed run. It should replace the shared proxy with a small open-weight decoder-only Transformer while retaining matched parameter, token, context, and inference budgets. The benchmark should add human-reviewed relation annotations, relation permutations, and independent semantic coverage judgments.

Phase 2 turns semantic coverage into an explicit measurement. The function K in Equation 2 should be decomposed into entity, relation, norm, exception, purpose, evidence, and effect coverage. The model should receive incomplete cases deliberately and be scored on detection, calibrated uncertainty, abstention, escalation, and safe candidate formation.

Phases 3–5 add structural attention, function-oriented routing, and normative late binding one at a time. Each phase should retain A1 and B controls, arbitrary-bias or random-router controls where applicable, and a fixed verifier beside the model ladder. A new mechanism should not be promoted because it increases rejection or refusal alone.

Phase 6 evaluates sequential domain and world-state updates with a frozen canonical axiology

Table 12: Phased experimental sequence.

Phase	Addition	Primary question	Gate
1	Small matched Transformer A1/B/C1/C2	Do the proxy effects persist under neural sequence modeling?	Capability, structural-OOD, calibration, security, and UCR thresholds
2	C3 semantic-coverage awareness	Can missing or contradictory context trigger calibrated abstention or escalation?	Coverage detection and false-confidence criteria
3	D structural attention	Does relation-aware attention improve transfer beyond context injection?	Relation-permutation and equal-compute controls
4	E structured routing	Do reasoning-function experts improve efficiency or specialization?	Routing audits, starvation tests, and compute-normalized gains
5	G normative late binding	Does governed runtime norm selection preserve authority and conflict state?	Context-swap, authority-spoofing, and action-path tests
6	Continual learning	Can world-state and domain updates occur without axiological or normative drift?	Fixed- O_A^v conformance and rollback tests
7	Formal assurance	Can candidate actions be compiled and checked with CSL-Core/Z3, TLA+, or an equivalent independently reviewed stack?	Invariant coverage, unknown-state handling, and enforcement binding

and explicit normative versioning. It should measure representation drift, relation forgetting, behavioral regressions, source provenance, compatibility, and rollback. Phase 7 formalizes the action compiler and assurance boundary; it should report exactly which properties are checked, which assumptions are required, and which states remain unknown.

9 Conclusion

This paper proposes VGA as a layered alignment model: a governed canonical axiology is kept distinct from its derived neural representation, contextual norms are late-bound, purpose selects domain relevance, world state remains evidence-bearing, and candidate actions cross an independent assurance boundary. The reference design makes these distinctions explicit in the representation, training, runtime, and evaluation interfaces.

The executed pilot tests the first A1/B/C1/C2 mechanism gate with a shared MLP on a deterministic synthetic benchmark. Its generated Results section reports primary metrics, structural-OOD behavior, candidate conformance, coverage degradation, calibration, attacks, and hypothesis criteria. Those observations identify which feature groups changed the proxy’s behavior and where semantic occlusion and incomplete coverage remain failure modes.

The next step is a preregistered small-Transformer study with human-reviewed semantic relations, independent coverage judgments, structural holdouts, and the same capability and security controls. Only after that gate should structural attention, routing, late binding, continual learning, or formal assurance be evaluated as additional mechanisms.

A Appendix: terminology, notation, and benchmark design

A.1 Terminology and critical distinctions

Values $\neq$ duties; axiology $\neq$ norm; ontology $\neq$ world state; evidence $\neq$ ground truth; authority $\neq$ truth; recency $\neq$ authority; structural conditioning $\neq$ formal assurance; behavioral compliance

Table 13: Terms used consistently in the manuscript.

Term	Definition
VGA	Value-Grounded AI Alignment research framework.
VGTA	Value-Grounded Transformer Architecture reference design.
Canonical axiology O_A^v	External, typed, versioned, provenance-bearing, governed L0 artifact.
Axiological neural module A_ϕ^v	Derived learned or hybrid module; non-authoritative and linked to O_A^v .
Semantic-to-neural compilation	The governed relation $A_\phi^v = C_\phi(O_A^v)$.
Axiological conformance	Empirical relation recovery and behavior-probe property, not semantic proof.
Structural value conditioning	Proposed influence of value structure on neural representation, attention, routing, or state.
Normative authority	Metadata used to govern applicability; not a scalar moral weight.
Normative conflict	Explicit unresolved state for incompatible applicable assertions.
Normative late binding	Runtime selection of governed norms under context, authority, and provenance.
Domain ontology / purpose model	L2a possible domain structure / L2b task-specific relevance model.
World state	L3 time-indexed assertions with source, confidence, and provenance; not ground truth.
Extrinsic assurance	Independent action compilation, verification, and enforcement outside neural computation.

$\neq$ alignment proof; self-improvement $\neq$ authority to redefine values; care-oriented behavior $\neq$ subjective love.

A.2 Canonical and world-state notation

The canonical source is an external artifact O_A^v with a deployment-assigned content identifier. A derived module is $A_\phi^v = C_\phi(O_A^v)$, and A_ϕ^v cannot become authoritative by learning. An instance world-state assertion is

$$(s, p, o, t_s, t_e, \text{source}, \text{confidence}, \text{provenance}).$$

The tuple supports temporal validity and epistemic uncertainty. It does not turn confidence into truth or justify embedding volatile facts into model weights.

A.3 Benchmark matrix

A.4 Synthetic fixture equations

For an illustrative candidate action a , the repository’s toy scorer computes

$$S(a) = U_{\text{task}}(a) + \sum_{j=1}^m w_j f_j(a),$$

where U_{task} is task utility, f_j are supplied value features, and w_j are context-bound weights. The A1 control selects by U_{task} alone. The fixed fixture verifier checks a small set of hard predicates. These equations are transparent and limited; they are not a learned moral objective.

A.5 Falsification and implementation checklist

Future studies should preregister thresholds for meaningful capability-normalized gains, conformance, calibration, over-refusal, unsafe acceptance, bypass, semantic completeness, latency, and attack success. The framework is weakened if ontology conditioning reduces general reasoning, fails on structural OOD, merely memorizes graph relations, increases false confidence or over-refusal, creates unacceptable cost, increases attack success, collapses under coverage loss, or matches A1 only at higher complexity. A verifier result is not an alignment result.

Table 14: Proposed benchmark matrix for future empirical work.

Family	Perturbation	Primary metric	Required control
Representation	Relation deletion, permutation, and graph completion	Recovery, concept-direction stability	Random labels and matched text
Moral salience	Omit or paraphrase relevant relations	Patient/relation recall	A1 without structure
OOD alignment	Lexical, compositional, and structural splits	Accuracy, effect size, calibration	Overlap audit; structural OOD priority
Normative conflict	Authority, jurisdiction, exception, and stale-source swaps	Conflict recognition, abstention, escalation	No scalar-average baseline only
Purpose relevance	Purpose swap and irrelevant-subgraph injection	Useful relevance and over-refusal	Same domain with matched context
Prompt robustness	Paraphrase, injection, and jailbreak	Attack success and recovery	Matched attack budget
Coverage	100/75/50/25 percent, contradictory, malicious, irrelevant	Capability/alignment degradation, false confidence	Coverage-seeded replication
Assurance	Candidate and verifier perturbation	Rejection, unsafe acceptance, bypass, UCR	Same fixed verifier and action schema
Pluralism	Consensus, disagreement, authorized profiles	Distributional similarity, diversity, adherence	Preserve judgment distributions
Continual learning	Sequential domain and norm updates	Drift, conformance, regression	Frozen-module baseline

Before deployment of any future implementation, check: source version and digest; compiler reproducibility; conformance and drift gates; normative authority and conflict state; purpose and world-state provenance; action-schema completeness; fixed-verifier coverage; audit and rollback targets; and governance approval. The current repository satisfies only the interface and fixture portions of this checklist.

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